Continuing Sequences

Exam code: 0607

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Continuing sequences

How do I continue a given sequence?

- You can work out the **first differences** to see if there is a **pattern**
  - The first differences are the values the sequence changes by each time
- For example
  - $4, \quad 7, \quad 10, \quad 13$
    $(+3) \ (+3) \ (+3)$
    - The first differences are all +3
    - The next term is 13 + 3 = 16
  - $6, \quad 1, \quad -4, \quad -9$
    $(-5) \ (-5) \ (-5)$
    - The first differences are all -5
    - The next term is -9 - 5 = -14
  - $2, \quad 8, \quad 15, \quad 23$
    $(+6) \ (+7) \ (+8)$
    - The first differences increase by 1
    - The next term is 23 + 9 = 32
  - $1, \quad 3, \quad 7, \quad 15$
    $(+2) \ (+4) \ (+8)$
    - The first differences double each time
    - The next term is 15 + 16 = 31

Sequences of squares, cubes, and triangular numbers

- Sequences can often be formed using **square**, **cube**, or **triangular** numbers
- It can help to be familiar with these sequences of numbers
- **Square** numbers are the results of squaring integers
  - $1^2, 2^2, 3^2, 4^2, 5^2, 6^2, 7^2, 8^2, 9^2, 10^2, 11^2, 12^2, \ldots$
  - 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, ...
- **Cube** numbers are the results of cubing integers
  - $1^3, 2^3, 3^3, 4^3, 5^3, \ldots$
  - 1, 8, 27, 64, 125, ...
- **Triangular** numbers are the result of summing consecutive integers

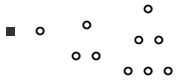
Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

triangles



Worked Example

Find the next term in the sequence

$$16,\ 19,\ 25,\ 37,\ 61$$

Answer:

Look at the first differences (the values the sequence changes by each time)

$$\begin{array}{ccccccccc} 16, & & 19, & & 25, & & 37, & & 61 \\ & (+3) & & (+6) & & (+12) & & (+24) & \end{array}$$

The differences are doubling each time, so the next difference will be $2 \times 24 = 48$

$$61 + 48$$

The next term is 109

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